Compensation in DC Government:

A Data-Driven Analysis

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MATH 014 — Introduction to Data Science
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Dataset: DC Public Employee Salary - DC Open Data

02 | Dataset Overview

40,476
Employees
active public workers

88
Agencies
government departments

$95,537
Median Salary
annual compensation

$413,088
Top Earner
highest individual salary

Key Variables

JOBTITLE
AGENCY (88 departments)
GRADE (pay band 1–16+)
COMP_RATE (salary or hourly)
HIRE_DATE (1966 – 2025)

Why it matters

DC's public salary data is a window into civic equity; who is rewarded, how much, and why. Analyzing it reveals structural patterns invisible from a single pay stub.

03 | Objective

Investigate how compensation is distributed across DC's public workforce; identifying which agencies, grades, titles, and employment types drive the highest and lowest pay, and how tenure correlates with earnings.

Q1

Which agencies employ the most workers, and how do their median salaries compare?

Q2

How does compensation scale across pay grades 1–16?

Q3

Which specific job titles command the highest median pay?

Q4

Does employee tenure predict annual salary?

Q5

Are salaried employees paid significantly more than hourly workers?

04 | Data Cleaning & Preparation

1. Rename & Standardize

Renamed columns for clarity (DESCRSHORT→AGENCY). Stripped whitespace, title-cased job titles.

2. Parse Dates

Converted HIRE_DATE strings to datetime objects. Computed TENURE_YEARS relative to 2025 snapshot.

3. Classify Pay Type

Flagged COMP_RATE < 200 as hourly, ≥ 1000 as salaried. Dropped 135 ambiguous records (<0.4%).

4. Annualize Salaries

Estimated hourly annual earnings at 2,080 hrs/year (40 hrs × 52 weeks) for fair comparison.

5. Handle Missing

Only GRADE had missing values (40 rows, 0.1%). Retained in dataset; excluded from grade-specific analyses.

6. Remove Duplicates

Confirmed zero duplicate rows in the raw dataset.

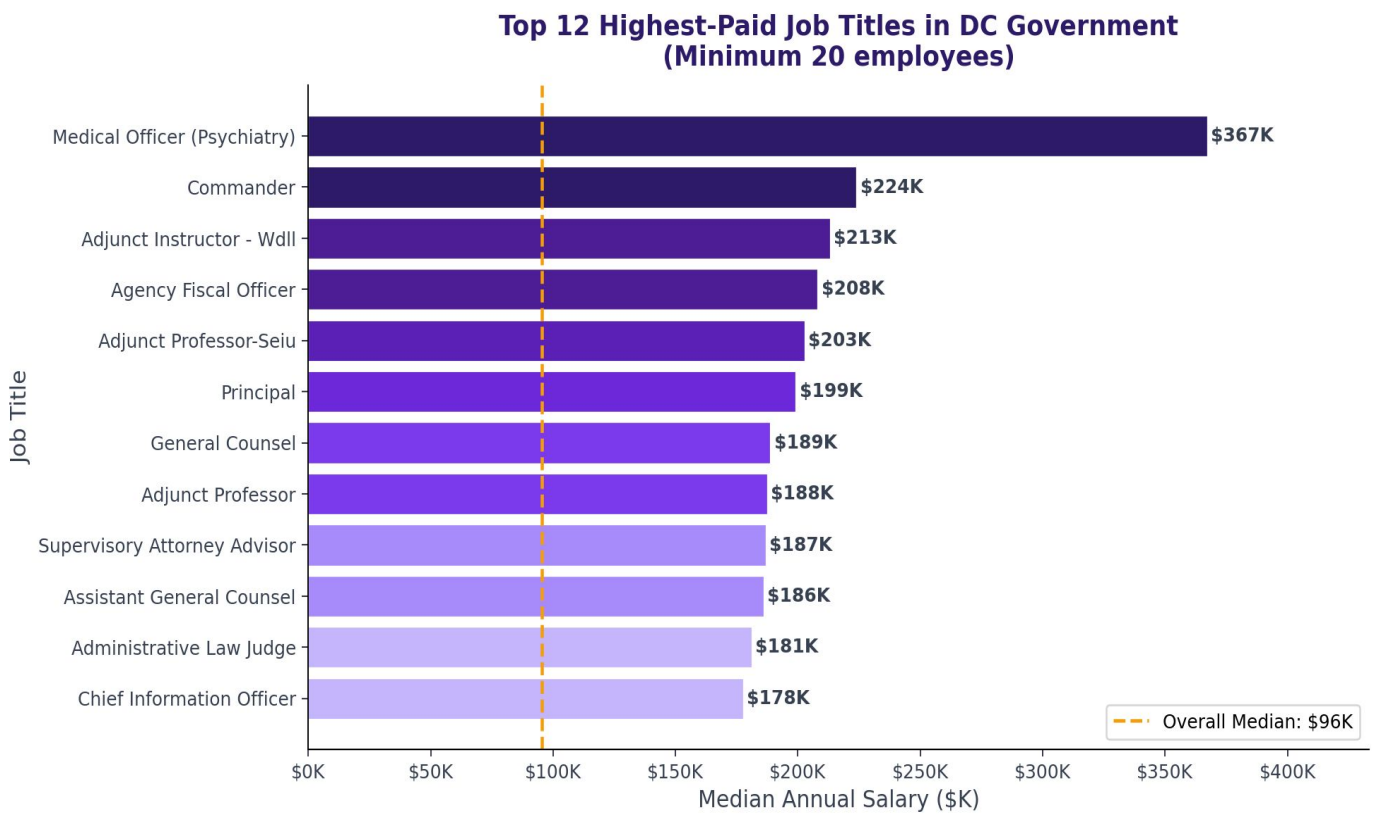
05 | Dataset Snapshot

First 6 rows of the cleaned dataset; each row is one DC government employee.

First Name	Last Name	Job Title	Agency	Grade	Annual Salary	Hire Year	Pay Type	Tenure (yrs)
Natasha	Boley	Compliance Specialist	DBH	13	$123,873	2004	Salaried	21
Korlu	Wolobah-Kuyon	Psychiatric Nurse	DBH	09	$115,907	2020	Salaried	5
Hadizat	Elayo-Ross	Supervisory Psychiatric Nurse	DBH	13	$122,500	2023	Salaried	2
Gary	Evans	Climate Control Maint Mechanic	DBH	11	$86,798	1985	Hourly	40
Jemia	Gross	Special Police Officer	DBH	08	$75,787	2016	Salaried	9
James	Broughton	Program Monitor	DBH	12	$107,046	1987	Salaried	38

9 columns shown · 40,476 total rows after cleaning · Names are real public record data from DC Open Data

06 | Visualization — Top-Paying Job Titles



$367K

Medical Officer (Psych.) — highest title

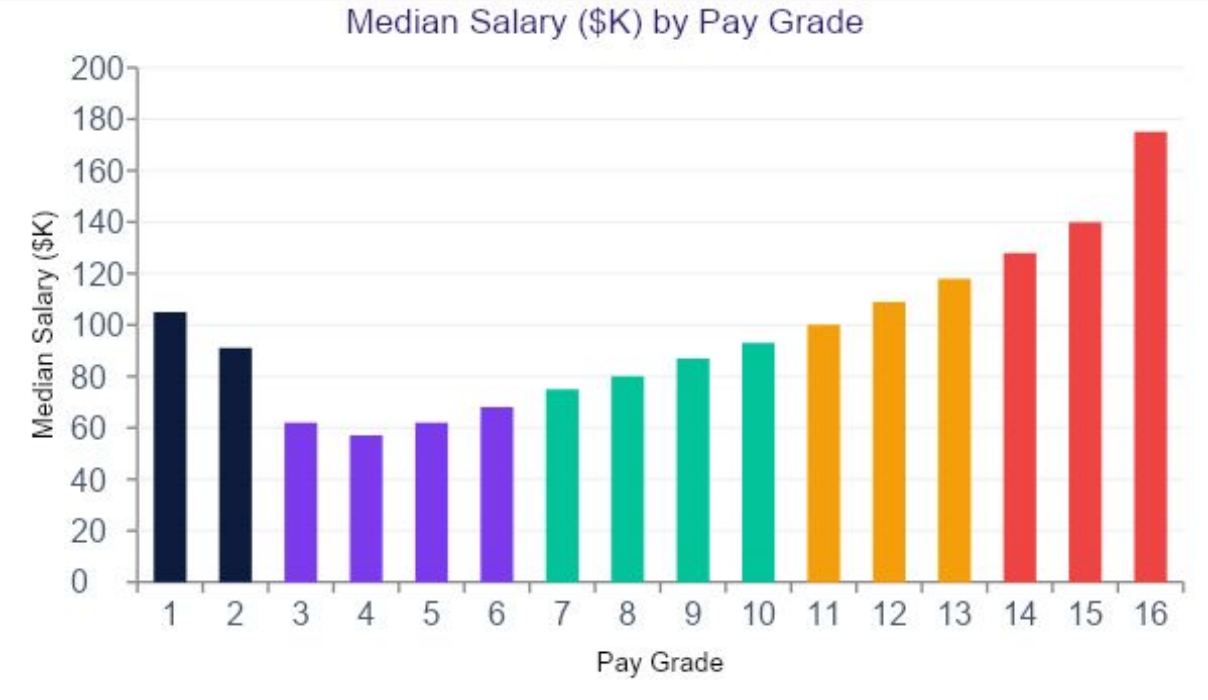
2×

top titles earn 2× the overall median

Legal

law & medicine dominate the top tier

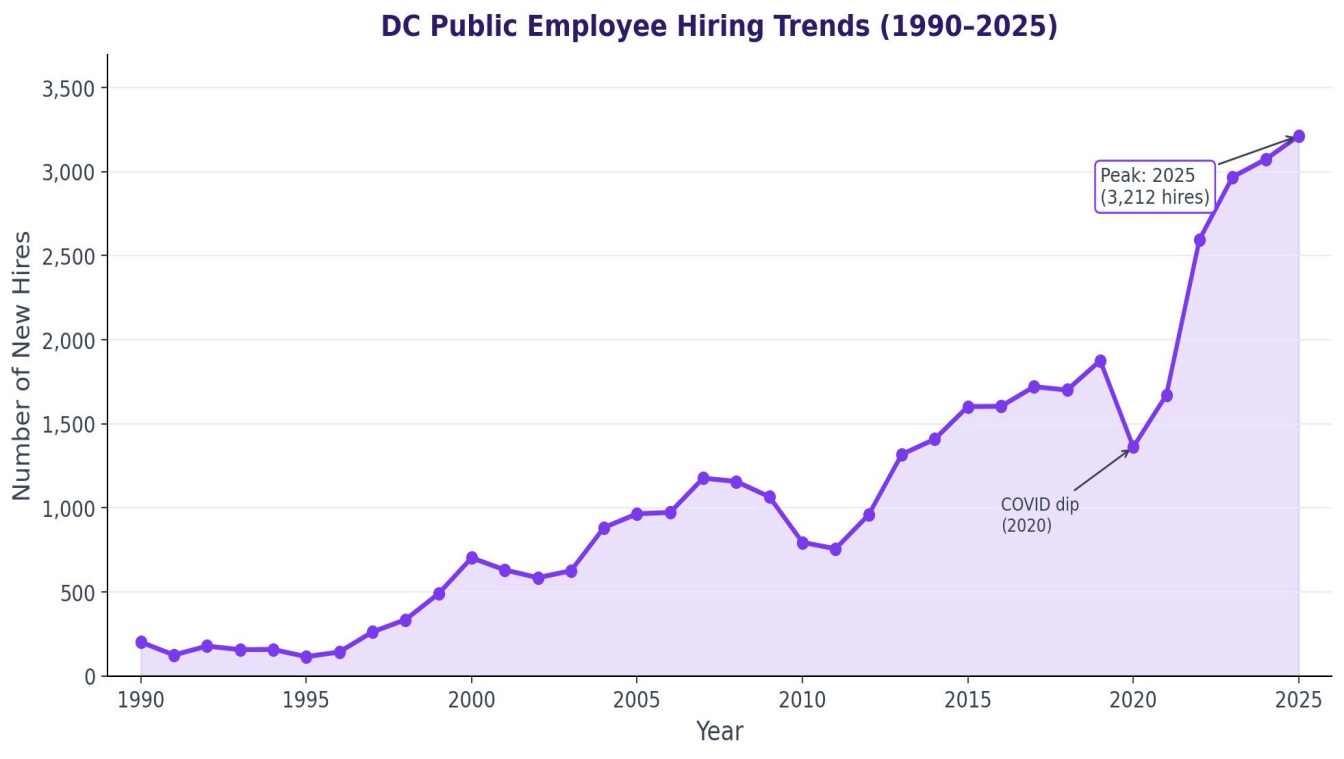
07 | Visualization — Salary by Pay Grade



Key Takeaways

- Clear upward trend: Grade 1→16 salary more than doubles
- Grades 14–16 accelerate sharply — leadership premium
- Grade 16 median: $175K — 1.7× Grade 1
- Pay grade is the single strongest structural predictor of salary

08 | Visualization — DC Hiring Trends



3,212

2025 Peak — record high

216/yr

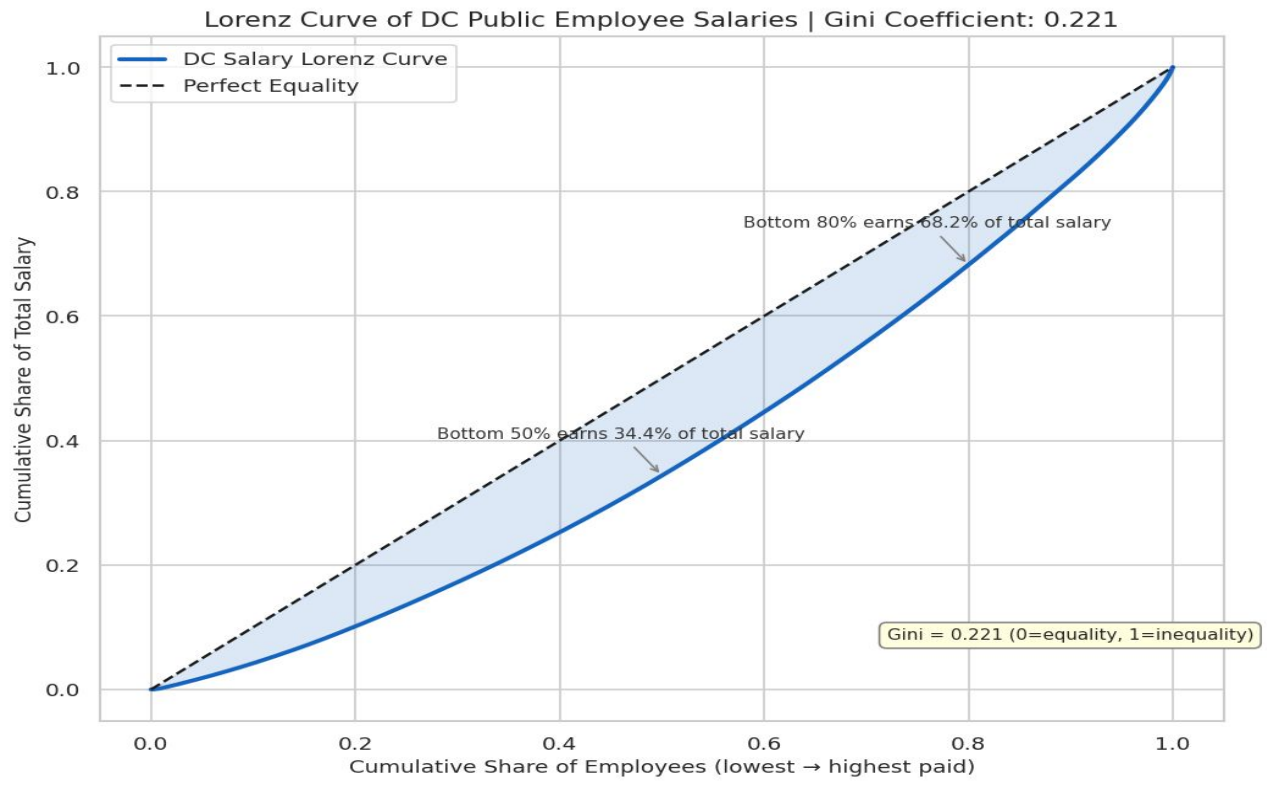
1990s avg hires per year

11×

growth vs. 30 years ago

Insight: Record hiring in 2022–2025 signals major expansion — but a large 1990s cohort now has 25–30 years of service and is approaching retirement, creating a succession risk.

09 | Surprise — The Lorenz Curve of DC Salaries



What is it?

Maps cumulative share of employees (x) vs. cumulative share of total salary (y). The further the curve bows below the diagonal, the more unequal the pay.

Gini = 0.221

moderate inequality

Bottom 50%: earns 25% of total salary

Top 20%: earns ~40% of total salary

vs. Private: US Gini ~0.45 — DC is 2× more equal

10 | Key Findings

01 Pay Grade Is the #1 Predictor

Grade 16 employees earn a $175K median vs. ~$57K at Grade 3–4. The grade ladder is the most reliable structural determinant of compensation.

02 Agency Composition Drives Pay Gaps

Specialized agencies (PSC, OCTO, HBX) have median salaries 45%+ above DCPS. It's not just the agency — it's who they hire.

03 A $46K Pay-Type Divide

Salaried workers earn a $100K median vs. $54K estimated for hourly workers — a structural two-tier workforce within a unified government.

04 Moderate Inequality (Gini 0.22)

DC's public payroll is more equal than the private sector (Gini ~0.45), but the bottom 50% still earns only 25% of total salary.

05 Surge in Recent Hiring

Record hiring in 2024–2025 signals workforce expansion — but a large 1990s-era cohort approaching retirement poses succession risk.

The Takeaway

In DC government, your pay grade, your agency, and your appointment type determine your salary far more than how long you've worked there. The workforce is more equal than the private sector; but structural divides still exist, particularly between hourly and salaried employees.

Grade First

Pay grade explains more salary variance than any other factor.

Specialize

Specialized agencies pay a consistent premium over large service agencies.

Type Matters

$46K gap between hourly and salaried — employment type is structural, not incidental.

Future work: merge demographic data for equity analysis | model salary prediction | track annual changes